

Q-R PRACTICAL EXAM - SCENARIO 1

Legend:

ESWB



Applied Blue Group Lock



Temporary Grounds



WARNING

SHOCK & ARC FLASH HAZARD

NO LOCAL MAIN CONSIDERED FOR
ARC FLASH CALCULATIONS.
UPSTREAM PROTECTIVE DEVICE APPLIES.

7' 2"

ARC FLASH BOUNDARY

5.1 cal/cm²

CALCULATED INCIDENT ENERGY
AT 36" WORKING DISTANCE

13200 V

Limited Approach Boundary: 5 ft. 0 in.
Restricted Approach Boundary: 2 ft. 2 in.

Study Rev. 2
Date Issued: 6-2025

Location: MVS-1



WARNING

SHOCK & ARC FLASH HAZARD



Location: ITL Chemical Corporation - Magnum DS
Report #: TQTQCOEXAMP.1 Rev. 2 Issued: JUN-2025

8' 0"

ARC FLASH BOUNDARY

7.9 cal/cm²

CALCULATED INCIDENT ENERGY
AT 24" WORKING DISTANCE

5' 1"

ARC FLASH BOUNDARY

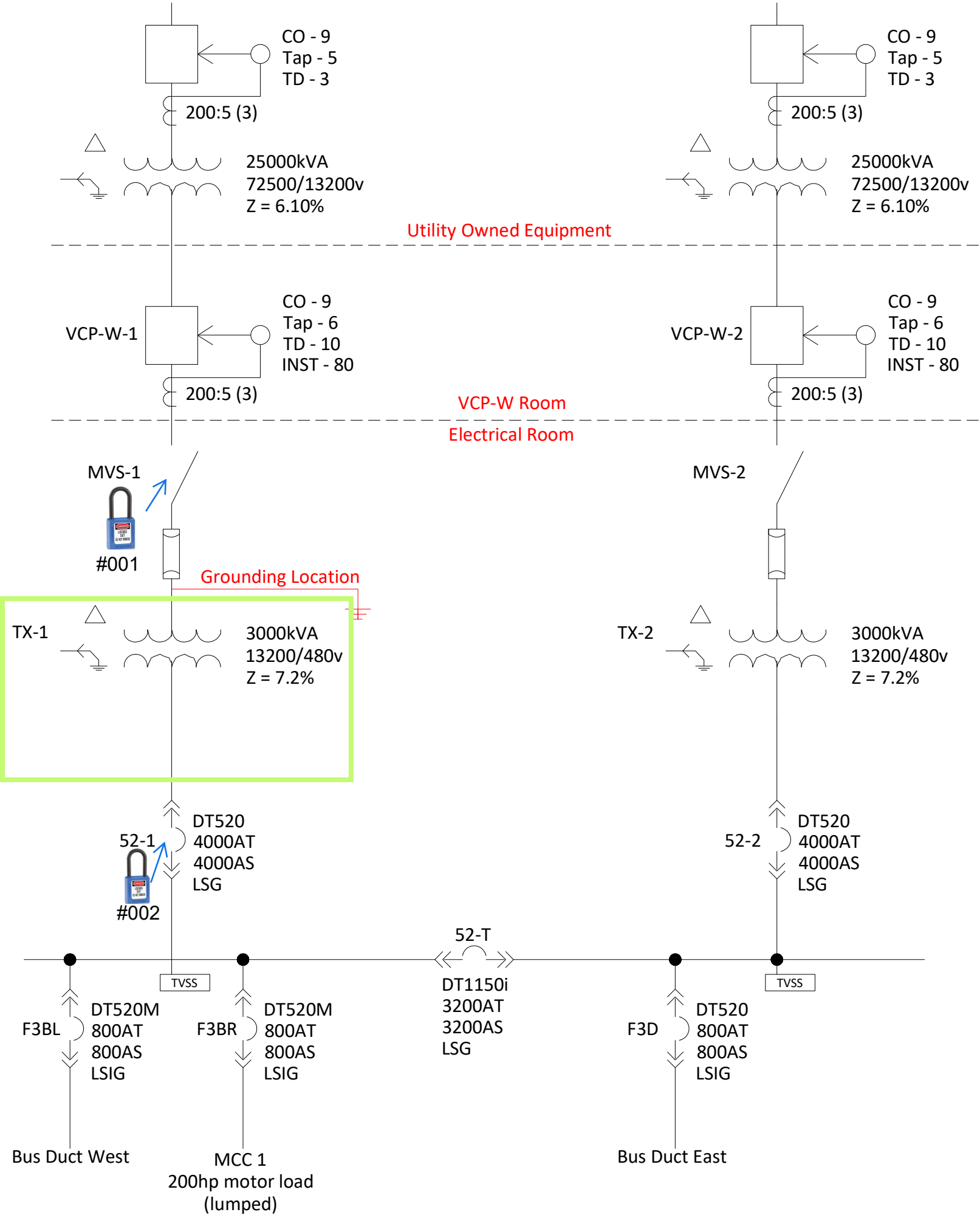
4.3 cal/cm²

CALCULATED INCIDENT ENERGY
AT 24" WORKING DISTANCE

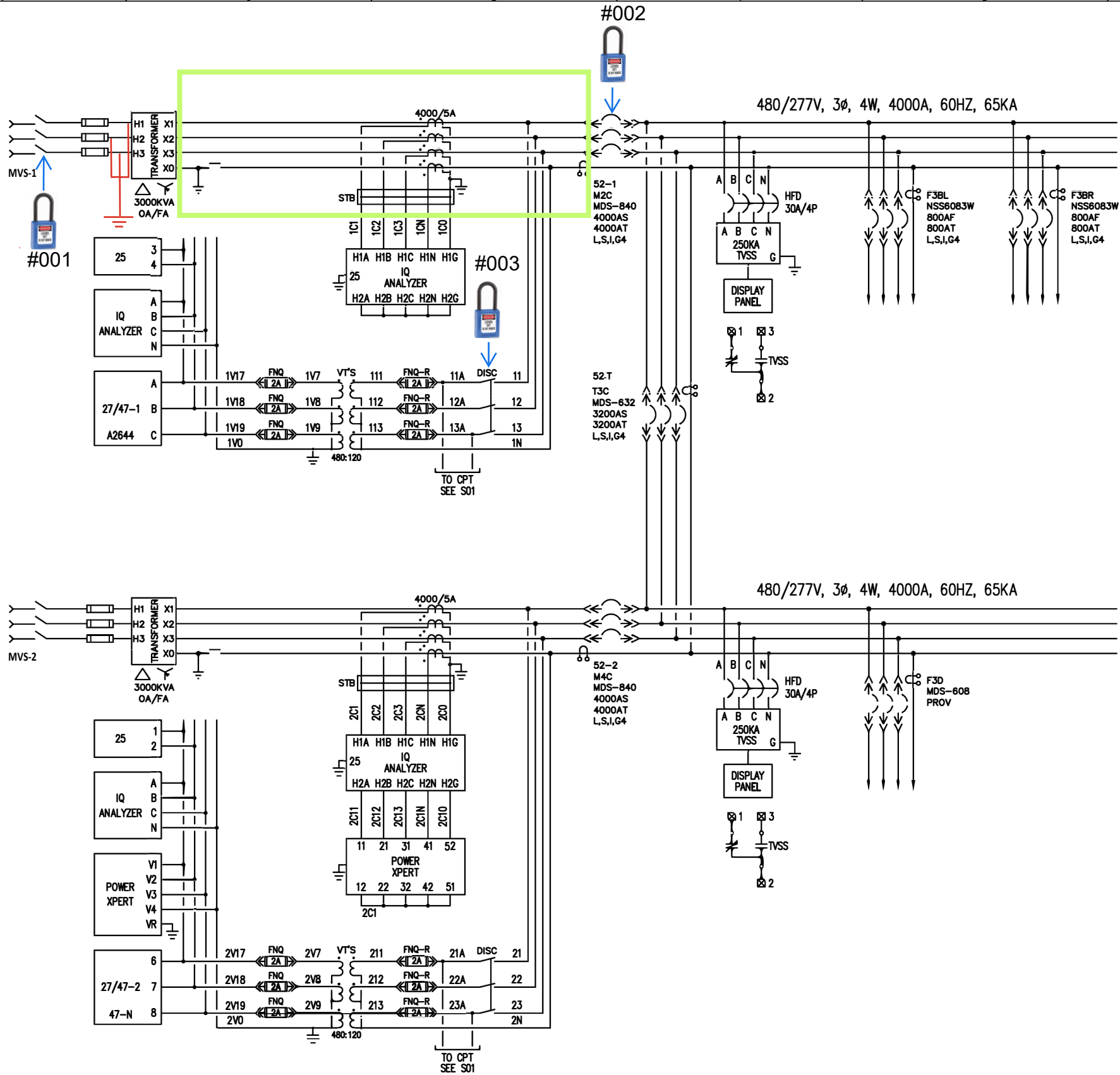
480 V Shock Hazard

Min. Glove Class: 00

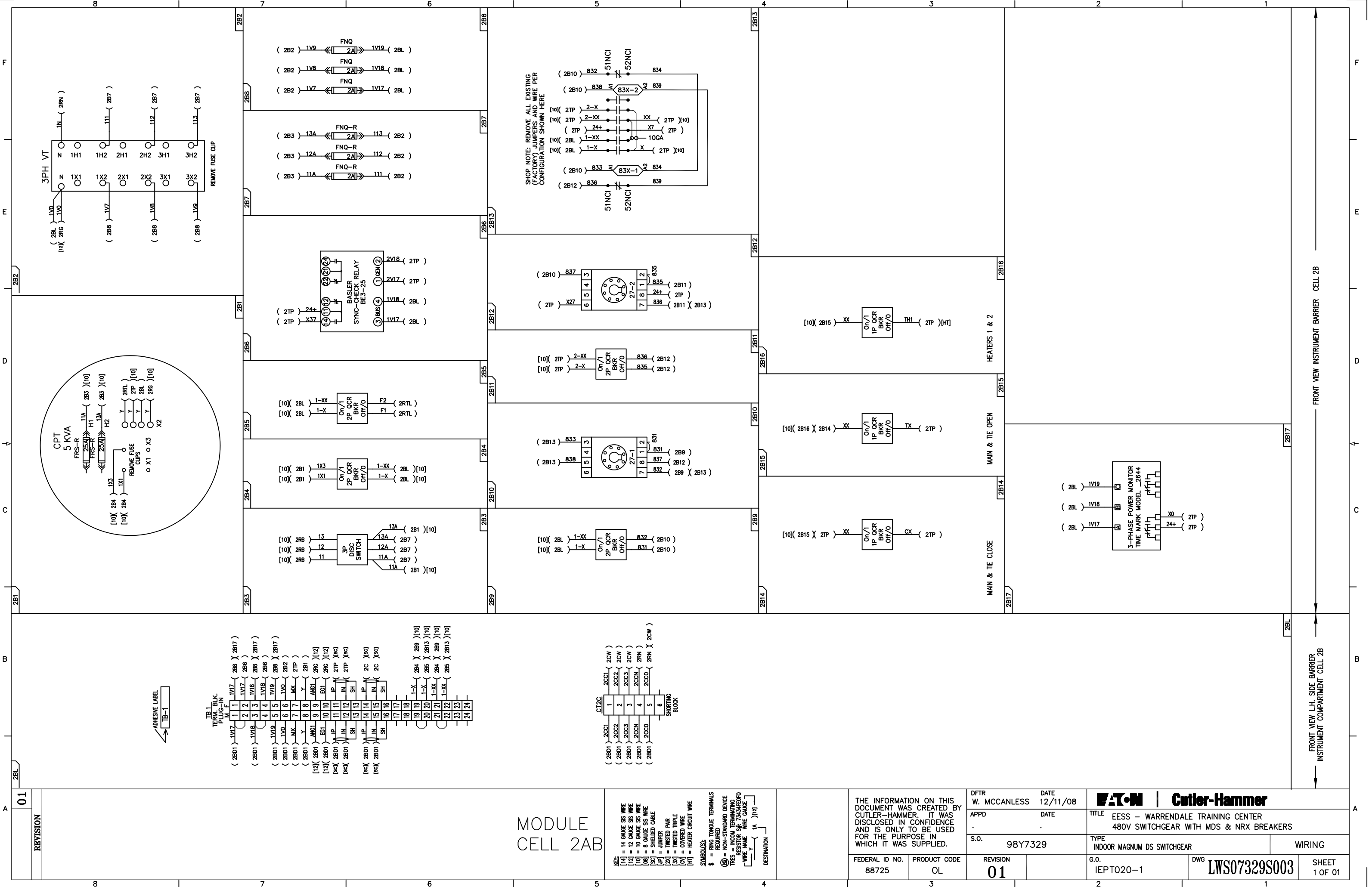
Limited Approach Boundary: 3' - 6"
Restricted Approach Boundary: 1' - 0"



Single Line Diagram of
Magnum DS Switchgear with
Arc Flash Labels



01	ISSUED FOR PROD	REVISION	02	WIRE NAMES	W. McCAULESS 12/17/08	03	AS BUILT, REVISED PIN OUTS FOR 25 DEVICE	B. HALL 02/20/09	THE INFORMATION ON THIS DOCUMENT WAS CREATED BY CUTLER-HAMMER. IT WAS DISCLOSED IN CONFIDENCE AND IS ONLY TO BE USED FOR THE PURPOSE IN WHICH IT WAS SUPPLIED.	DFTR SIMON		DATE 09/19/08		A-T-N Cutler-Hammer			
										APPD		DATE		TITLE EESS - WARRENDALE TRAINING CENTER 480V SWITCHGEAR W/MDS & NRX BREAKERS			
										S.O.		98Y7329		TYPE MAGNUM DS SWITCHGEAR		THREE LINE	
										FEDERAL ID NO. 88725		PRODUCT CODE CL		REVISION 03		G.O. IEPT020-1	



REVISION		DATE		TITLE		TYPE		WIRING	
01		W. MCCANLESS	12/11/08	APPD		INDOOR MAGNUM DS SWITCHGEAR			
		S.O.	98Y7329						
		FEDERAL ID NO.	88725	PRODUCT CODE	OL	G.O.	IEPT020-1	DWG	LWS07329S003
				REVISION	01				SHEET 1 OF 01

MODULE
CELL 2AB

KEY:
[14] = 14 GAUGE SIS WIRE
[12] = 12 GAUGE SIS WIRE
[10] = 10 GAUGE SIS WIRE
[08] = 8 GAUGE SIS WIRE
[SC] = SHIELDED CABLE
[JP] = JUMPER
[2W] = TWISTED PAIR
[3W] = TWISTED TRIPLE
[CV] = COVERED WIRE
[HT] = HEATER CIRCUIT WIRE

SYMBOLS:
\$ = RING TONGUE TERMINALS REQUIRED
Ⓢ = NON-STANDARD DEVICE
TRES = INCOM TERMINATING RESISTOR S.F. 754SHFEUQ
WIRE NAME WIRE GAUGE
DESTINATION

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DFTR W. MCCANLESS DATE 12/11/08

APPD DATE

S.O. 98Y7329

REVISION 01

G.O. IEPT020-1

DWG LWS07329S003

SHEET 1 OF 01

